

POWER BI PROJECT REPORT

Airport Analysis Dashboard

Flight volume, cancellations and delays across U.S. airports

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Microsoft Power BI · Interactive reporting

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Project overview

38K	453	60.72	34.99
Total flights	Cancelled flights	Avg arrival delay	Avg departure delay

This project is an interactive **Power BI** dashboard built to monitor key **airline performance metrics**: flight volume, cancellations and delays. It helps airline and airport teams spot patterns, understand where operations slow down and make data-driven decisions to improve punctuality and customer experience.

Objectives

- Track flight volume and cancellations by month, day, origin and destination.
- Measure average departure and arrival delays and how they relate to distance.
- Identify the routes and destinations most affected by cancellations.
- Give stakeholders a clear, interactive view of operational performance.

Dashboard structure

Page	Purpose	Main visuals
1. Airport Analysis Dashboard	Flight volume and cancellations	KPI cards, bar chart by month, area chart by day
2. Delays and Time Analysis	Punctuality of departures and arrivals	KPI cards, scatter plots by distance, map
3. Detailed Flight Analysis	Drill-down into individual flights	Detailed table, treemap of cancellations

Note: figures in this report are read from the dashboard as filtered in the screenshots. They change when slicers are applied.

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Page 1 – Airport Analysis Dashboard



Airport Analysis Dashboard (Power BI)

The first page gives a high-level view of activity. Two KPI cards show **38K total flights** and **453 cancelled flights**. Slicers for month, day, origin and destination filter every visual on the page.

Visuals

- **Total Flights by Month:** volume is steady at roughly 3.2K–3.3K flights per month, with February the lowest at 3.0K.
- **Cancelled Flights by Day:** cancellations are irregular across the month, with peaks of about 20 flights on certain days.

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Page 2 – Delays and Time Analysis



Delays and Time Analysis (Power BI)

The second page focuses on punctuality. The average **arrival delay (60.72)** is noticeably higher than the average **departure delay (34.99)** in the data shown. Slicers for month, year, origin and destination allow comparisons across periods and routes.

Visuals

- **Avg Dep Delay / Avg Arr Delay by Distance:** scatter plots showing how delays vary with flight distance.
- **Map:** departure and arrival delays by origin and destination airport across North America.

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Page 3 – Detailed Flight Analysis



Detailed Flight Analysis (Power BI)

The third page lets users drill down to individual flights and see where cancellations concentrate.

Visuals

- **Detailed table:** flight date, origin, destination, expected vs. actual departure and arrival times, cancellation flag and distance.
- **Treemap:** cancellations by destination. In the view shown, **Jackson (Mississippi)** has the largest block, followed by Atlanta, Los Angeles and Denver.

Key observations and next steps

Key observations

- Monthly flight volume is stable, with February the lowest (3.0K vs. about 3.2K–3.3K).
- Cancellations peak on certain days of the month rather than following a steady pattern.
- Arrival delays are higher than departure delays, suggesting delay is added in flight or on arrival.
- Cancellations are concentrated in a few destinations, led by Jackson (Mississippi) in the view shown.

Questions worth investigating

Observation	Follow-up question
Cancellation peaks on specific days	Do the peaks match weather events, holidays or particular airlines?
Arrival delay above departure delay	Is the gap linked to distance, congestion at destination or specific routes?
Jackson (Mississippi) leads cancellations	Is this driven by flight volume at that airport or by a higher cancellation rate?

Tools and skills

Area	Details
Tool	Microsoft Power BI Desktop
Data preparation	Power Query (cleaning and transformation)
Calculations	DAX measures (e.g. average delays, totals)
Data modeling	Building tables for filtering across pages
Visuals	KPI cards, bar chart, area chart, scatter plots, map, treemap, detailed table
Interactivity	Slicers (month, day, year, origin, destination) and page navigation buttons

Possible improvements

- Analyze delays and cancellations by airline and by cause.
- Add year-over-year comparisons.
- Publish the report to Power BI Service for online sharing.